

Analyzing Music Preferences and Audio Features Using Spotify Data

In this project, I aim to analyze the relationship between music listening habits and song characteristics using Spotify data. The goal is to understand how audio features such as energy, tempo, valence, and danceability influence listening preferences.

The data will be collected from the Spotify Web API and my personal Spotify listening history. The API will be used to retrieve detailed audio features of tracks, while my listening history will provide user-level behavioral data. Additionally, I may enrich the dataset by including popularity metrics and genre information obtained from Spotify.

The dataset is expected to include several hundred to thousands of songs, depending on the available listening history. Each sample will represent a track and include features such as artist, genre, tempo, energy, valence, and listening frequency.

Using this dataset, I plan to perform exploratory data analysis (EDA), statistical testing, and potentially machine learning techniques to identify patterns in music preferences. This project will provide insights into how musical features relate to user behavior.